Vasana Manas

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.17
Narayana Junior College	Class XII (CBSE)	2023	94.20%
Narayana School	Class X (CBSE)	2021	76.40%

PROJECTS

Time-Series Database for Financial Data

- Developed a high throughput, single node time series database, integrating write ahead log, lock free read Memtable, custom SSTable with block indexing and Bloom filters, achieving 1.28M writes/sec.
- Implemented crash safe recovery using CRC32C-checksummed WAL logs with automatic corrupt write detection, parsing and replaying 1M records back into the MemTable in 913 ms.
- Optimized query performance using Bloom filters, and an LRU block cache to minimize disk reads, and structured the MemTable as an L-level Skip List to accelerate point lookups to an expected O(log N) time complexity.

Multithreaded Full Text Search Engine

- Built a multithreaded C++ search engine with a custom Trie inverted index. Implemented a hand-rolled thread pool using a staged fork-join design, achieving a theoretical 3.9× speedup on 8 cores.
- Designed a BM25 relevance scoring pipeline with a custom bounded max-heap storing parallel score and ID arrays, extracting top-K from 5,000 candidates in 24.9 µs.
- Used sorted vectors with n interval skip pointers for posting lists achieving AND query intersections in $O(n + m)$.

Portfolio Optimization via Cumulative Prospect Theory | PyTorch, Python

Group Project

- Formulated portfolio allocation under Cumulative Prospect Theory and built Minorization-Maximization and Convex-Concave Procedure solvers to handle non-concave objectives.
- Developed a GPU-parallel Projected Gradient Ascent solver using PyTorch autodiff, scaling optimization to 10× larger sample sizes than traditional convex QP methods.
- Evaluated parameter sensitivity and initialization strategies, showing that using mean-variance starting allocations consistently accelerated solver convergence.

Attendance Tracking System | ReactJS, ExpressJS, SimpleWebAuthn, JWT, NodeMailer

- Built a full-stack attendance platform with JWT role based access control and dual layer proxy verification via Haversine geofencing and WebAuthn based device binding.
- Developed attendance analytics dashboards with class wise trends, attendance history visualizations with charts and a Nodemailer service for automated low attendance alerts.

SKILLS

- **Languages:** C,C++,Javascript,SQL
- **Web Development:** HTML, CSS, ReactJS, NodeJs, ExpressJs, Web Authn, JWT, Websockets
- **Databases:** MySQL, PostgreSQL, Redis

RELEVANT COURSES

- **Mathematics:** Probability and Random Process, Matrix theory, Calculus, Differential Equations, Convex optimization, Pattern Recognition and Machine Learning, Vector Calculus, Complex Variables
- **Computer Science:** Data Structures and Applications, Database Management Systems, Programming in C,

Operating Systems
- **Communication:** Signals and Systems, Digital Signal Processing, Communication Systems, Wireless Communication

POSITIONS OF RESPONSIBILITY

Web Dev Team Lead | EML IITH Club

- Led a 5-member team to develop and manage web applications supporting club operations and outreach.

Web Dev Team Lead | NSS IITH Club

- Maintained the PostgreSQL database for NSS IITH to support daily operations and student data management.

Technical Coordinator | OCS IITH Club

- Collaborated with a team to maintain the OCS website and database

EXTRACURRICULAR

- Won a gold medal in inter-hostel and inter-departmental cricket tournaments.
- Volunteered at the NSS Vidhyadhan event to teach school students.